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# Step 1: Import necessary libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Step 2: Create dataset manually (instead of CSV)
data = pd.DataFrame({
    'Head Size(cm^3)': [4512, 3738, 4261, 3777, 4177, 3585, 3785, 3559, 3613, 3982],
    'Brain Weight(grams)': [1530, 1297, 1335, 1282, 1590, 1300, 1400, 1250, 1350, 1450]
})

print("Dataset:\n", data)

# Step 3: Prepare the data
x = np.array(data['Head Size(cm^3)'])
y = np.array(data['Brain Weight(grams)'])

# Step 4 & Step 5: Calculate mean, slope (m), intercept (b)
def calculate_coefficients(x, y):
    x_mean = np.mean(x)
    y_mean = np.mean(y)

    # Calculate slope (m)
    numerator = np.sum((x - x_mean) * (y - y_mean))
    denominator = np.sum((x - x_mean) ** 2)

    m = numerator / denominator

    # Calculate intercept (b)
    b = y_mean - (m * x_mean)

    return m, b

m, b = calculate_coefficients(x, y)

print("\nSlope (m):", m)
print("Intercept (b):", b)

# Step 6: Make predictions
y_pred = m * x + b

# Step 7: Plot the regression line
plt.scatter(x, y, label="Data Points")
plt.plot(x, y_pred, label="Regression Line")

plt.xlabel("Head Size(cm^3)")
plt.ylabel("Brain Weight(grams)")
plt.title("Simple Linear Regression (Least Squares)")
plt.legend()
plt.show()

# Step 8: Calculate R-squared value
def calculate_r2(x, y, m, b):
    y_mean = np.mean(y)

    # Total Sum of Squares
    TSS = np.sum((y - y_mean) ** 2)

    # Residual Sum of Squares
    RSS = np.sum((y - (m * x + b)) ** 2)

    # R-squared
    R2 = 1 - (RSS / TSS)

    return R2

r2 = calculate_r2(x, y, m, b)
```

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# Step 9: Display results
print("\nR-squared value:", r2)

# Step 10: Using sklearn (in-built method)
from sklearn.linear_model import LinearRegression

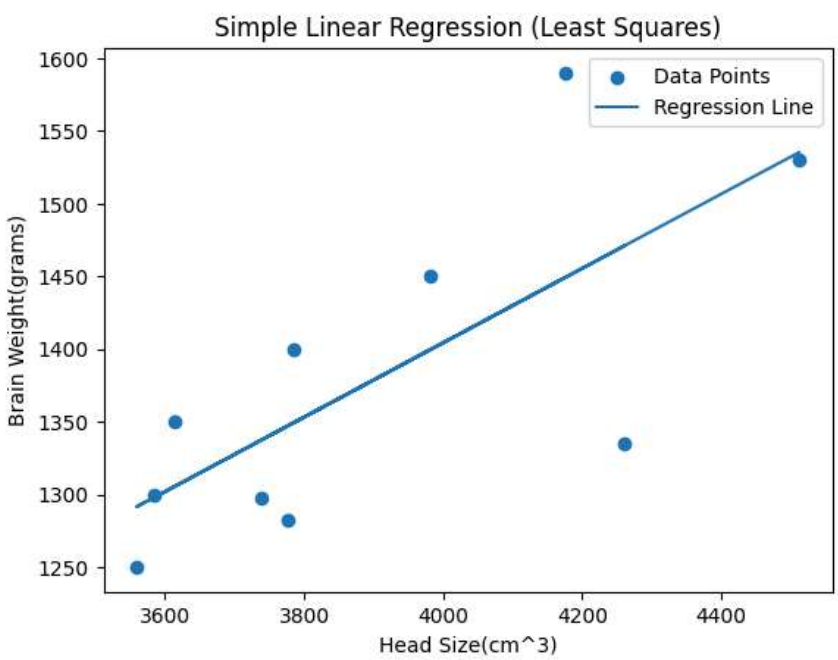
x_reshaped = x.reshape(-1, 1)

model = LinearRegression()
model.fit(x_reshaped, y)

print("\n--- Using sklearn ---")
print("Slope (m):", model.coef_[0])
print("Intercept (b):", model.intercept_)
print("R-squared value:", model.score(x_reshaped, y))
```

```
Dataset:
    Head Size(cm^3)  Brain Weight(grams)
0              4512                 1530
1              3738                 1297
2              4261                 1335
3              3777                 1282
4              4177                 1590
5              3585                 1300
6              3785                 1400
7              3559                 1250
8              3613                 1350
9              3982                 1450

Slope (m): 0.25594472835221965
Intercept (b): 380.4970986275308
```

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R-squared value: 0.5378041273560665

--- Using sklearn ---
Slope (m): 0.25594472835221954
Intercept (b): 380.4970986275313
R-squared value: 0.5378041273560665
```